

Shay Mordechai

Security Researcher | Vulnerability Discovery & Protocol Analysis

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Summary

Security Researcher with a proven track record of discovering Zero-Days (3 pending CVEs) and building custom Python tooling. Expert in protocol analysis and processing large-scale data (1TB+ daily logs). Currently advancing expertise in Malware Analysis, Threat Detection, and Incident Response methodologies.

Professional Experience

DevSecOps Intern | Israel Tax Authority | Aug 2024 – Jul 2025

- Spearheaded Secure SDLC and cloud migration to Checkmarx One, mitigating SQLi and DOM XSS across 15+ pipelines.
- Performed massive data manipulation on 1TB+ of daily logs to identify anomalous behaviors and drive actionable Incident Response.

Combat Soldier | Combat Intelligence Collection Corps, IDF (Hesder) | Aug 2016 – Apr 2018

Executed covert reconnaissance and visual anomaly detection to build actionable operational target profiles.

Education

B.Sc. Computer Science | Jerusalem College of Technology | 2021-2025 | GPA: 89

Coursework: Network Analysis (97), Reverse Engineering (93), Information Security (90).

Continuous Learning: Advanced Malware Analysis (In Progress) – Mastering static/dynamic analysis, IoCs extraction, unpacking, and custom Yara rules for threat detection.

Technical Projects & Research *(Details in Portfolio)*

Vulnerability Research – SBCL Compiler (3 Zero-Days / CVEs)

- Managed full-cycle research via Python fuzzing, uncovering 3 Zero-Days (CVEs pending). Authored technical documentation for MITRE and National CERT (#11265).

Network Protocol & Browser Security Research

- Discovered DNS Rebinding attack vectors threatening local networks; proposed deterministic TTL-hardening mitigations validated by TP-Link PSIRT.
- Published analysis on RWX memory allocation & CFI bypasses in V8 JIT browser engines.

TLS 1.2 Handshake & PCAP Simulator

- Engineered a 4,000+ line **Python-based tool** simulating TLS handshakes; used **Wireshark** and **tcpdump** to dissect complex **TCP/IP** flows and generate SSLKeyLogs.

Automated Research Engine ("Predator") - Developed automated security solutions from scratch, integrating AI tools to process data and detect anomalous patterns in real-time.

Technical Skills

- **Security:** Network Traffic Analysis, Vulnerability Discovery, Malware Analysis, Yara Rules, Incident Response, Fuzzing.
- **Dev & Data:** Python (Custom Tooling), SQL, Bash, Data Processing.
- **Networking:** Wireshark, Tshark, TCP/IP, Scapy, Raw Sockets, DNS Rebinding, AWS.